Agents That Remember

Designing Memory-Driven AI Systems

Rakesh Lakshminarayana

May 2026

Around the world from February till June 2026

AgentCamp

Global AI Bootcamp 2026

https://github.com/rakeshL4

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Session Agenda

- **Foundations of Context Engineering**
  - Why Context Matters for Agents
  - Understanding Agent Memory Architecture
- **Hands-on Demo**
  - Long-Term Memory: Personalization
  - Episodic Memory: Recall Conversations
- **Q&A**

Why Context Matters

LLMs are Stateless

LLMs have no built-in memory. Context is essential to maintain continuity and coherence across multi-turn conversations.

Bring Your Own Data

Ground responses with your domain-specific, and up-to-date knowledge.

Drives Response Quality

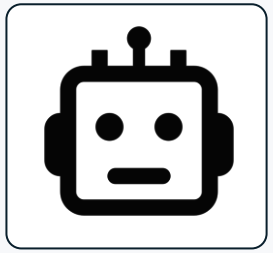
Improve quality and relevance of responses by providing the right context

Personalization

Unlocks personalized interactions by remembering user preferences, tone, and behavioral patterns

What forms the Context ?

Context engineering is designing, structuring, and managing the information sent to a language model to improve responses.



AI Agent

System Instructions - Persona

System prompt, rules, constraints, and few-shot examples

Knowledge - Grounding

Docs, databases (RAG), past interactions, relevant memory

Tools & Skills - Action Layer

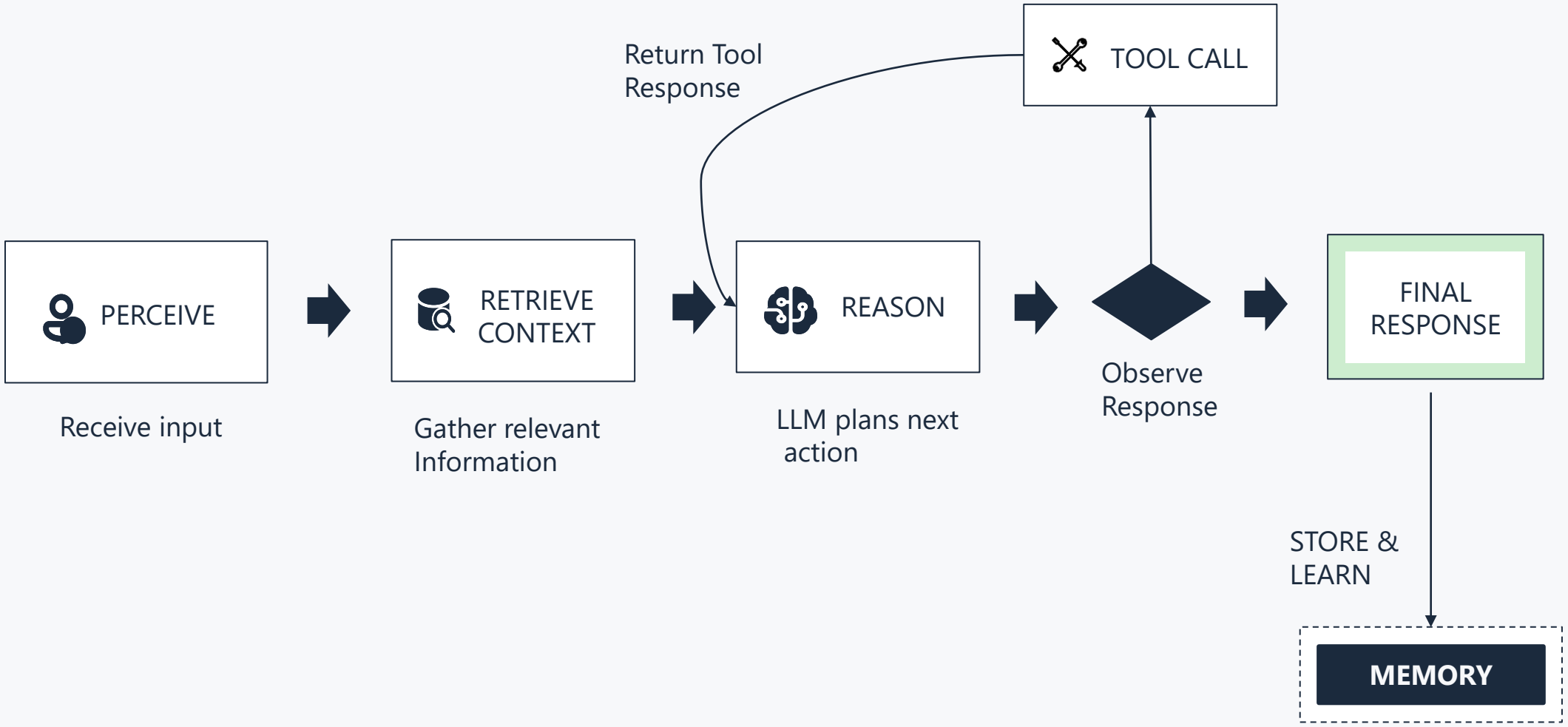
Tool schemas and descriptions

Current State

Chat history, intermediate reasoning, tool traces

Large Language Model

Agent Loop - Understanding How Agents Run



Types of Memory in Agents

Short-Term
Working Memory

What's happening now

Holds the current conversation and immediate context, allowing the agent to maintain flow and respond coherently within a session.

Chat history, tool results

Semantic
Knowledge Memory

What the agent knows

General domain knowledge retrieved via RAG from documents, databases, and knowledge bases.

Docs, policies, expertise

Episodic
Experience Memory

What it's done before

Records of past interactions and outcomes allowing the agent to learn from previous experiences and reuse them when relevant

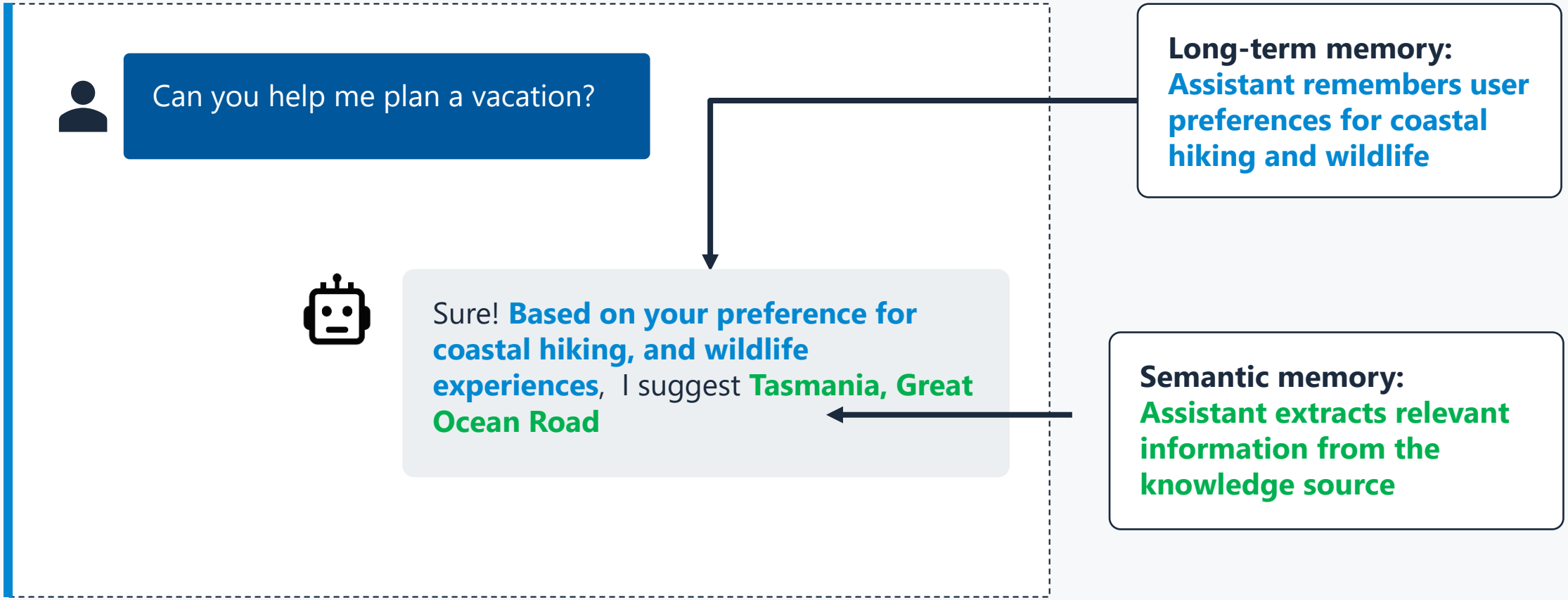
Past conversations, and events

Long-Term
Persistent Memory

What the agent remembers

Stores information across sessions, such as user preferences, and learned facts, enabling personalization over time.

User prefs, past decisions


Can you help me plan a vacation?
Sure! Based on your preference for coastal hiking, and wildlife experiences, I suggest Tasmania, Great Ocean Road
Long-term memory:
Assistant remembers user preferences for coastal hiking and wildlife
Semantic memory:
Assistant extracts relevant information from the knowledge source

What was the place I visited last year with great hiking?

You went to the Blue Mountains, Australia.

Do you want to try something similar this time?

Episodic memory:
Assistant retrieves past conversation about Blue Mountains trip

Short-term memory:
Assistant keeps track of current conversation about planning a new trip

Workshop

Designing Memory-Driven AI Agents

Designing Memory-Driven AI Systems
Welcome
Environment Setup
Lab 1 - Short-Term Memory
Lab 2 - Long-Term Memory
Lab 3 - Semantic Memory
Finishing Up
Learning Resources

Agents That Remember: Designing Memory-Driven AI Systems

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GITHUB REPOSITORY

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Prerequisites
What You'll Build
Architecture Overview
Getting Started

Session Overview

[Preview] README.md - agentcamp-workshop [Codespaces: bug-free guide] - Visual Studio Code

https://bug-free-guide-7vpv7jrvwrpf4x9.github.dev

agentcamp-workshop [Codespaces: bug-free guide]

EXPLORER

AGENTCAMP-WORKSHOP [CODESPAC...

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- azure.yaml
- package-lock.json
- pyproject.toml
- README.md

OUTLINE

TIMELINE

SOLUTION EXPLORER

[Preview] README.md

AI Travel Assistant Workshop

This repository contains the code and documentation for building an AI Travel Assistant using Microsoft Azure Framework.

For complete setup instructions and lab exercises, please visit:

https://binarytrails-ai.github.io/agentcamp-workshop/

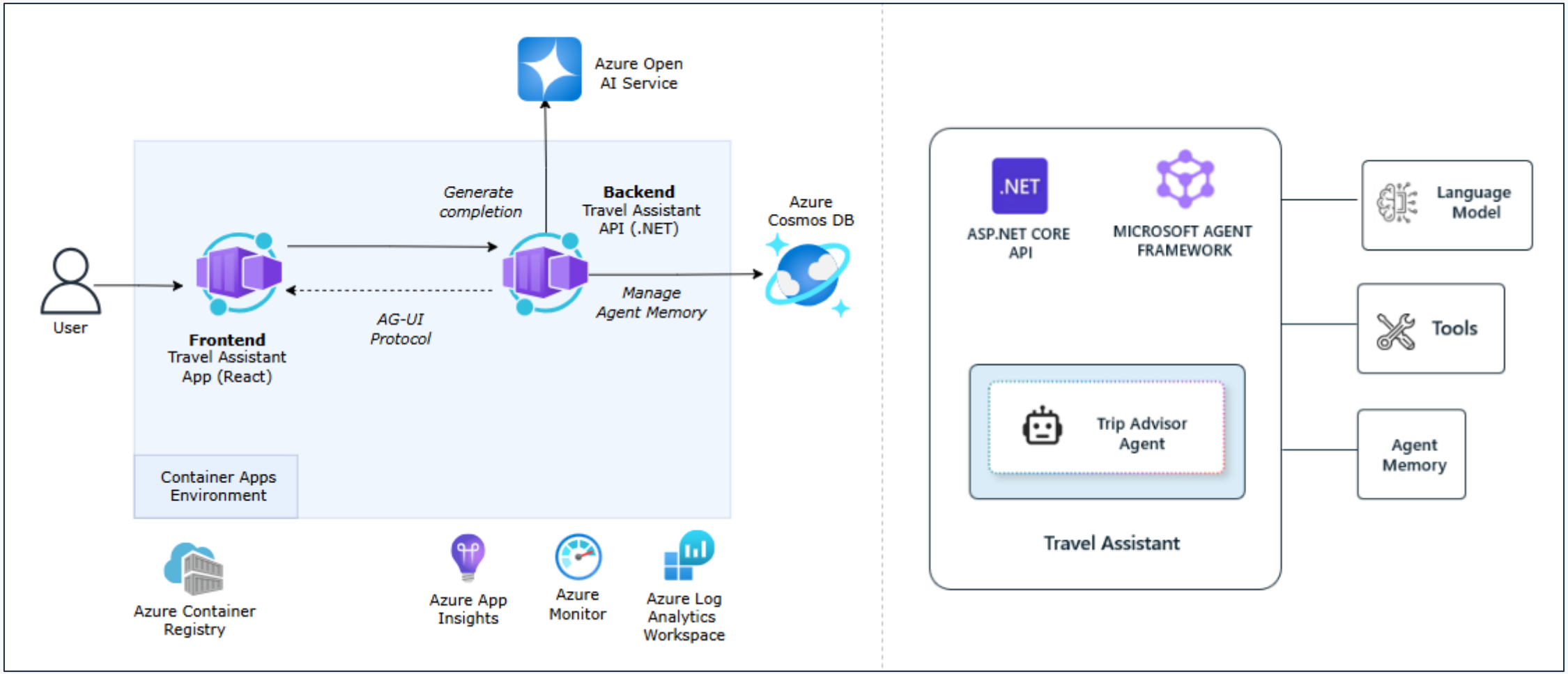
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE POLYGLOT NOTEBOOK

bash

```
@rakeshl4 ➜ /workspaces/agentcamp-workshop (main) $
```

Codespaces: bug-free guide | main* | 0 0 | 0 | Projects: 3 | Debug Any CPU | Solution file opened: agentcamp-workshop.sln | Finish Setup | No Environment | Layout: US

Travel Assistant – Architecture



DEMO
Short-Term Memory

[Preview] README.md - agentcamp-workshop [Codespaces: fuzzy potato] - Visual Studio Code

https://fuzzy-potato-97j76gq7575cp4qw.github.dev

Chat

agentcamp-workshop [Codespaces: fuzzy potato]

EXPLORER

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```
@rakeshl4 ➜ /workspaces/agentcamp-workshop/src/ContosoTravel.AppHost (main) $
```

bash frontend

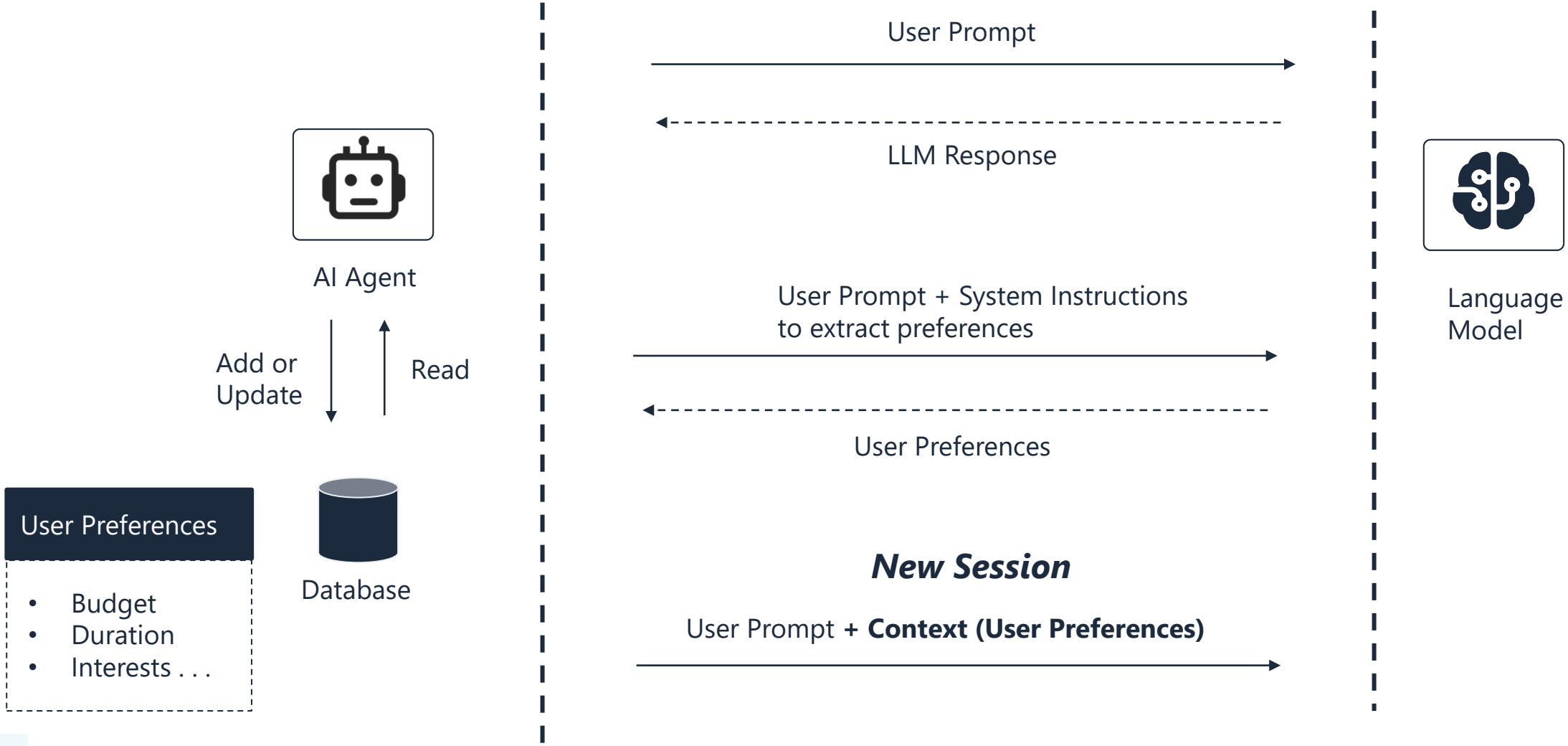
bash Contos...

Codespaces: fuzzy potato | main | 0 0 | 0 | Projects: 3 | Debug Any CPU | Solution file opened: agentcamp-workshop.sln | Finish Setup | Layout: US

DEMO

Personalization using Long-Term Memory

Personalized Experience using Long-Term Memory



ContosoTravelAgentBuilder.cs - agentcamp-workshop [Codespaces: fuzzy potato] - Visual Studio Code

https://fuzzy-potato-97j76gq7575cp4qw.github.dev

EXPLORER

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OUTLINE

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SOLUTION EXPLORER

[Preview] README.md | ContosoTravelAgentBuilder.cs 2, M

src > backend > Agents > ContosoTravelAgentBuilder.cs > { } ContosoTravelAgent.Host.Agents > ContosoTravelAgentBuilder > CreateAsync() : Task<AIAgent>

```
9   namespace ContosoTravelAgent.Host.Agents;
128     public Task<AIAgent> CreateAsync()
129     {
130         // Get userId from HttpContext (fallback to threadId or default-user)
```

PROBLEMS 2 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS 1 | AZURE | POLYGLOT NOTEBOOK

```
^C@rakeshl4 ➜ /workspaces/agentcamp-workshop/src/ContosoTravel.AppHost (main) $ dotnet run
/workspaces/agentcamp-workshop/src/backend/Tools/DestinationSearchTool.cs(152,19): warning CS1591: Missing XML comment for publicly visible type o
r member 'DestinationResult.Description'
/workspaces/agentcamp-workshop/src/backend/Tools/DestinationSearchTool.cs(155,25): warning CS1591: Missing XML comment for publicly visible type o
r member 'DestinationResult.Experiences'
/workspaces/agentcamp-workshop/src/backend/Tools/DestinationSearchTool.cs(158,19): warning CS1591: Missing XML comment for publicly visible type o
r member 'DestinationResult.SimilarityScore'
info: Aspire.Hosting.DistributedApplication[0]
      Aspire version: 13.2.2+25961cf7043e413abaf8ad84348988f2904b90d5
info: Aspire.Hosting.DistributedApplication[0]
      Distributed application starting.
info: Aspire.Hosting.DistributedApplication[0]
      Application host directory is: /workspaces/agentcamp-workshop/src/ContosoTravel.AppHost
warn: Microsoft.AspNetCore.DataProtection.Repositories.FileSystemXmlRepository[60]
      Storing keys in a directory '/home/vscode/.aspnet/DataProtection-Keys' that may not be persisted outside of the container. Protected data wi
ll be unavailable when container is destroyed. For more information go to https://aka.ms/aspnet/dataprotectionwarning
info: Aspire.Hosting.DistributedApplication[0]
      Now listening on: http://fuzzy-potato-97j76gq7575cp4qw-15160.app.github.dev
info: Aspire.Hosting.DistributedApplication[0]
      Distributed application started. Press Ctrl+C to shut down.
info: Aspire.Hosting.Devcontainers.DevcontainerSettingsWriter[0]
      Port forwarding (aspire-dashboard-http): http://localhost:15160
info: Aspire.Hosting.Devcontainers.DevcontainerSettingsWriter[0]
      Port forwarding (backend-http): http://localhost:5001
info: Aspire.Hosting.Devcontainers.DevcontainerSettingsWriter[0]
      Port forwarding (aspire-dashboard-otlp-grpc): http://localhost:21063
info: Aspire.Hosting.Devcontainers.DevcontainerSettingsWriter[0]
      Port forwarding (frontend-http): http://localhost:3000
^C@rakeshl4 ➜ /workspaces/agentcamp-workshop/src/ContosoTravel.AppHost (main) $ ^C
@rakeshl4 ➜ /workspaces/agentcamp-workshop/src/ContosoTravel.AppHost (main) $
```

bash frontend
bash Contos...
bash Contos...

Codespaces: fuzzy potato | main* | 0 2 | 1 | Projects: ContosoTravelAgent.Host | Debug Any CPU | Solution file opened: agentcamp-workshop.sln | Spaces: 4 | UTF-8 with BOM | LF | { } C# | Finish Setup | Layout: US

DEMO

Recall Conversation with Episodic Memory

ContosoTravelAgentBuilder.cs - agentcamp-workshop [Codespaces: fuzzy potato] - Visual Studio Code

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src > backend > Agents > ContosoTravelAgentBuilder.cs > { } ContosoTravelAgent.Host.Agents > ContosoTravelAgentBuilder > CreateAsync() : Task<AIAgent>

```
9   namespace ContosoTravelAgent.Host.Agents;
138             ?? "default-user";
139
140         var logger = _loggerFactory.CreateLogger<ContosoTravelAgentBuilder>();
141         AIAgent agent = _chatClient.AsAIAgent(new ChatClientAgentOptions
```

PROBLEMS 2 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | AZURE | POLYGLOT NOTEBOOK

bash frontend
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```
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 1.15.2 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-mr8r-92fq-pj8p
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```

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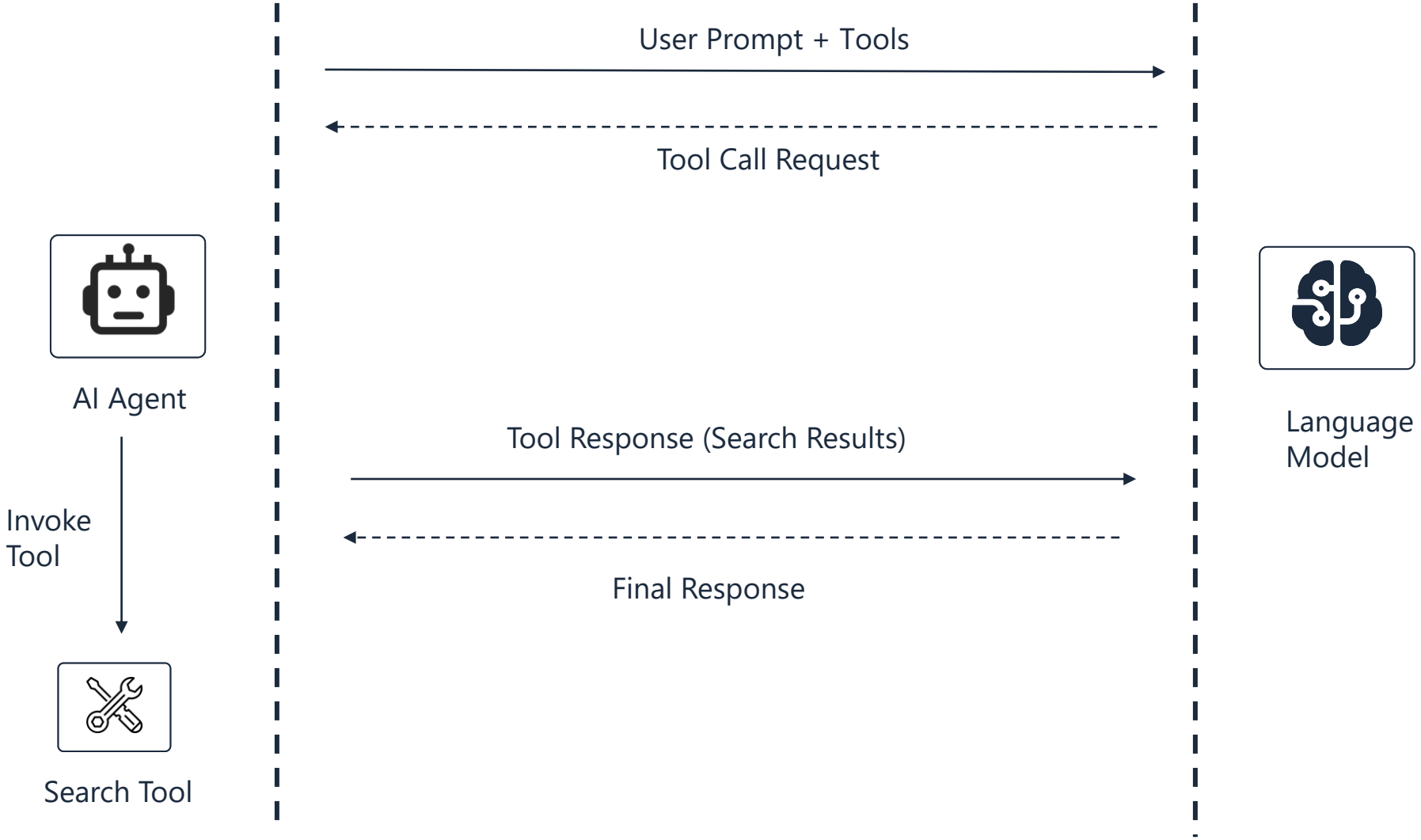
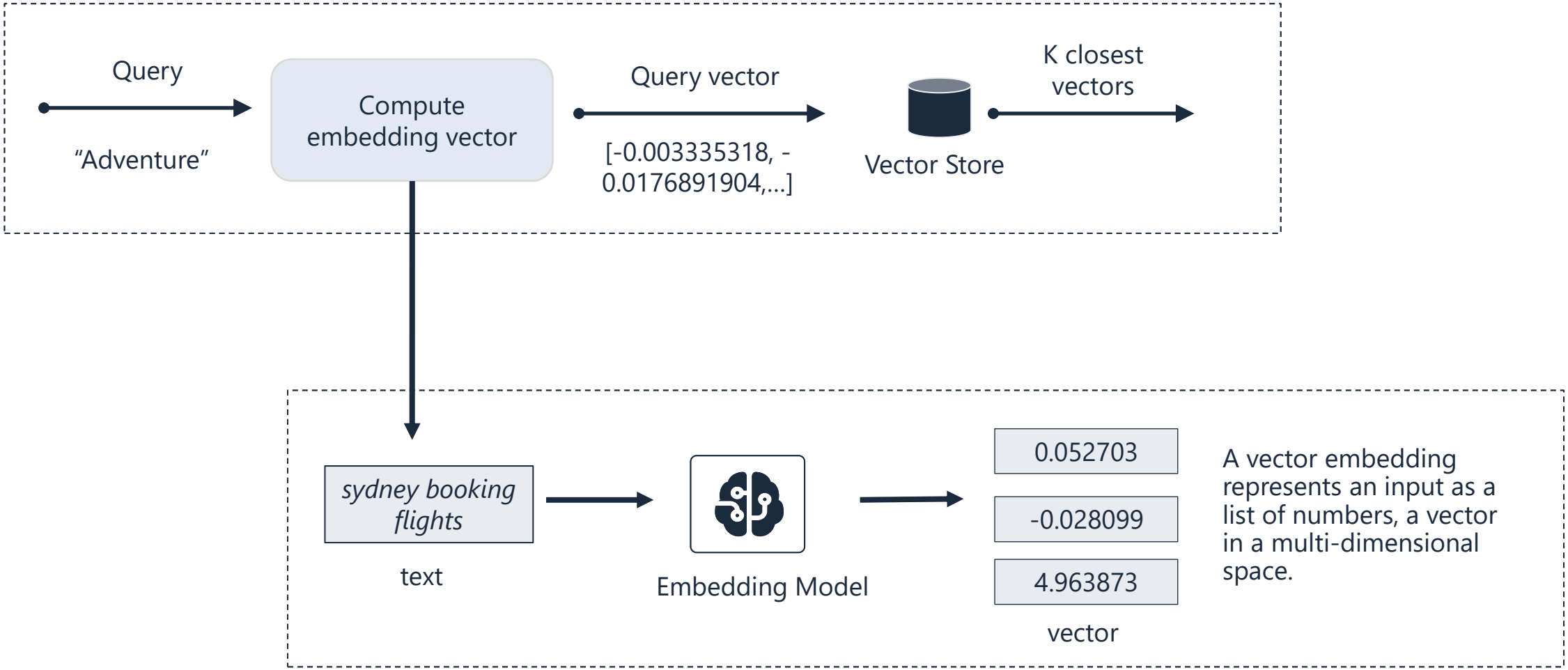
Recalling Past Interactions using Episodic Memory



Vector Search



DEMO
Query Planning

ChatHistorySearchTool.cs - agentcamp-workshop [Codespaces: fuzzy potato] - Visual Studio Code

https://fuzzy-potato-97j76gq7575cp4qw.github.dev

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      - UserContextTools.cs
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SOLUTION EXPLORER

[Preview] README.md | ContosoTravelAgentBuilder.cs 2, M | ChatHistorySearchTool.cs M

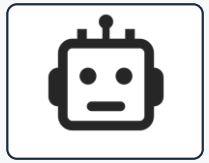
src > backend > Tools > ChatHistorySearchTool.cs > { } ContosoTravelAgent.Host.Tools > ChatHistorySearchTool > SearchChatHistory(string, string?) : Task<string>

EnrichQueryAsync Aa ab .* 1 of 1

```
7   namespace ContosoTravelAgent.Host.Tools;
13  public class ChatHistorySearchTool
41      private async Task<string> EnrichQueryAsync(string query)
67      }
68
69      /// <summary>
70      /// Searches past conversations to recall previous discussions, bookings, and user preferences
71      /// </summary>
72      [Description("Search past conversations to recall previous discussions, bookings, preferences, and interactions. Use when user ref
        1 reference
73      public async Task<string> SearchChatHistory(
74          [Description("Natural language query describing what to search for in past conversations (e.g., 'flights', 'hotel bookings',
75          [Description("Optional: Filter by specific user ID to search only that user's conversation history")] string? userId = null)
76      {
77          try
78          {
79              _logger.LogInformation("[ChatHistorySearch] Searching chat history for query: {Query}, userId: {UserId}", query, userId ?
80
81              var container = _database.GetContainer(_containerName);
82              // Generate embedding for the search query
83              var embeddingResponse = await _embeddingClient.GenerateEmbeddingAsync(query);
84              var queryEmbedding = embeddingResponse.Value.ToFloats().ToArray();
85
86              // Build WHERE clause with optional user filter
87              var whereClause = "c.ApplicationId = @appId";
88              if (!string.IsNullOrEmpty(userId))
89              {
90                  whereClause += " AND c.UserId = @userId";
91              }
92
93              // Use Cosmos DB vector search with VectorDistance function
94              var queryText = $@"
95                  SELECT TOP 10
96                      c.id,
97                      c.UserId,
```

Codespaces: fuzzy potato | main* | 0 2 | 10 | Projects: ContosoTravelAgent.Host | Debug Any CPU | Solution file opened: agentcamp-workshop.sln | Spaces: 4 | UTF-8 | LF | C# | Finish Setup | Layout: US

Agentic RAG



1 Query Understanding & Planning

- Analyze request
- Select knowledge sources
- Create retrieval plan

2 Iterative Retrieval

- Execute staged retrievals
- Vector Database
- Tools & APIs
- Internet

3 Synthesis Information

- Combine all sources
- Apply constraints
- Resolve conflicts

4 Self-Evaluation

Enough Context ?

- ❌ NO → More retrievals
- ✅ YES → Final response

↻ Loop if needed

I want to visit New Zealand for hiking.

Milford Track
Duration: 3-4 days
Difficulty: Moderate to Difficult
Highlights: Fjordland scenery, waterfalls

Routeburn Track
Duration: 2-3 days
Difficulty: Moderate
Highlights: Alpine passes, mountain views

AGENT'S ANALYSIS

Intent: Plan hiking trip to New Zealand

AGENT'S PLAN

- ✓ Retrieve NZ hiking destinations from knowledge base
- ✓ Ask user about constraints
- ✓ Get current trail conditions via tools
- ✓ Synthesize personalized recommendation

Next steps

1. Learn more about Microsoft Agent Framework
   aka.ms/agent-framework

2. Build and deploy Agents on Microsoft Azure
   https://learn.microsoft.com/en-us/training/paths/develop-ai-agents-on-azure/

3. AI Agent samples using Microsoft Agent Framework
   github.com/microsoft/agent-framework

4. Learn more about Agent–User Interaction protocol
   https://docs.ag-ui.com/

THANK YOU !

Let's stay connected

AI Agent Workshop

Please share your Feedback

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